

Queueing-Based Rate Allocation and Routing in Small Networks

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Introduction

- Packet-switched networks forward information as packets through routers/servers; queues form whenever arrivals exceed service for short periods.
- Queue buildup directly impacts key performance metrics: stability, mean backlog (queue length), delay, and (with finite buffers) packet drops.
- This project validates three canonical queueing models (Kendall notation): M/M/1, D/D/1, and finite-buffer M/M/1/N, by comparing simulated steady-state metrics with theory.
- We also plot transient behavior (time evolution of backlog and per-packet delay) to show how systems approach steady state.
- Finally, we extend the idea to a small multi-hop network and test a queue-informed routing/rate-allocation heuristic against simple baselines.
- **Main takeaway:** single-queue theory matches simulation when assumptions are met, and queue-aware decisions in small networks can reduce bottleneck congestion and improve end-to-end delay/loss under high load.

Methodology

- **Simulator:** Python discrete-event simulator on a directed graph. Each link has service rate μ (pkt/s) and each node uses FIFO queues. Flows inject packets at sources and packets are forwarded hop-by-hop to the destination.
- **Queue-aware routing:** estimate the recent arrival rate λ_e into each outgoing link queue and assigns a congestion cost $\delta e = 1/(\mu_e - \lambda_e + \epsilon)$. Next hops are chosen by minimizing the remaining path cost (shortest path using δe as edge weights).
- **Rate split:** when multiple outgoing links are usable, split traffic inversely with congestion: $x_{f,e} \propto 1/\delta e$, then normalize so total assigned rate on a link does not exceed μ_e .
- **Baselines:** (i) equal splitting, (ii) static shortest-hop routing (no queue awareness).

Results

A. Single-queue validation

In isolated single-queue experiments, the simulator reproduces the expected trends:

- For **M/M/1** with $\rho < 1$, measured $E[T]$ increases sharply as $\rho \rightarrow 1$, matching $E[T] = 1/(\mu - \lambda)$.
- For **M/M/1/N**, finite buffering caps delay at the expense of nonzero p_N ; as offered load increases beyond μ , throughput saturates near μ while drop probability rises.



- For **D/D/1** with $\lambda \leq \mu$, queueing delay is approximately zero and packets experience near-constant service time.

B. Multi-hop network evaluation

On small routed topologies with competing flows, queueing-informed decisions reduce the chance that one link becomes a persistent bottleneck. Compared to equal splitting and static routing, the queueing-based heuristic:

- decreases end-to-end delay at moderate-to-high offered load by diverting traffic from congested links,
- reduces drop probability when buffers are finite by preventing prolonged queue build-up at a single node,
- maintains comparable throughput, with improvements most noticeable when multiple feasible paths exist.

TABLE I
SMALL-NETWORK SIMULATION COMPARISON
($t_{\text{end}} = 3000\text{s}$, WARMUP=300S, SEED=1)

Policy	Thr. (pkt/s)	Drop	Delay (s)
Static SH	17.10	0.287	1.504
Queue-aware	19.64	0.183	1.343

Results

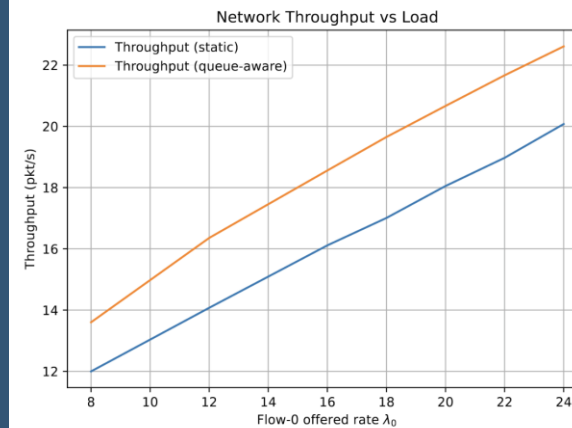


Fig: Aggregate throughput versus offered load λ_0

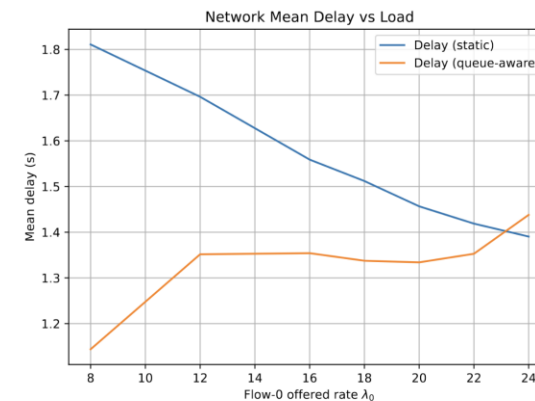


Fig: Mean delay (delivered packets) versus offered load λ_0

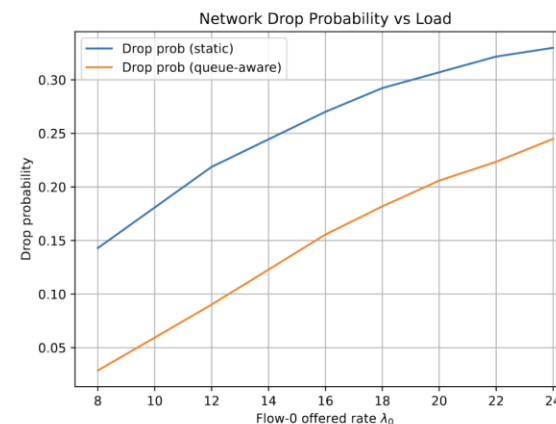


Fig: Drop probability versus offered load λ_0

Conclusion

This paper revisited three foundational queueing models (M/M/1, M/M/1/N, and D/D/1) and used their steady-state insights to guide a practical, queue-aware routing and rate allocation heuristic for small networks. Our discrete-event simulator validates the analytical single-queue behavior and shows that queue-informed control can improve end-to-end delay and loss relative to naive baselines when multiple routing/rate choices exist.

Future work includes:

1. Extending the heuristic to multi-commodity queue models with per-destination queues.
2. Integrating explicit control objectives (e.g., proportional fairness or delay minimization) with stability guarantees.
3. Evaluating on ns-3 or emulation testbeds to better match real protocol behaviors and wireless link dynamics.

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